



Operations Research

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Exercise Sheet 9

Training Exercises

Exercise T9.1. *Euler vs Hamilton*

For each of the tasks give a graph G with at least three vertices which fulfil the following properties:

- a) G is both Eulerian and Hamiltonian.
- b) G is neither Eulerian nor Hamiltonian.
- c) G is not Eulerian but Hamiltonian.
- d) G is Eulerian but not Hamiltonian.
- e) G does not represent a symmetric TSP instance.
- f) G represents a symmetric TSP instance that is not metric.

Exercise T9.2. *Filling of ATMs*

The bank *Total Untrustworthy Money (TUM)* maintains a number of ATMs for its retail customers at locations A, B, C, D, E, F, G around its headquarters in Garching-Hochbrück (H). Each of the ATMs must be refilled every seven days. For this purpose, the bank sends two employees from H with a small cash transporter to drive off the ATMs of the respective day and change the cash cassettes. The cost of a trip depends linearly on the distance traveled and the distances between the locations are known. Since the employees in question also have other tasks that need to be completed, they are only supposed to go out to fill the ATMs twice a week at the most. Also, no ATM can be visited multiple times within one week. The bank is now asking for a cost-minimizing solution.

- a) To which of the problem classes mentioned in the lecture within integer optimization (Assignment, Knapsack, Bin Packing, Set Covering, Traveling Salesperson) can the described situation be assigned? Give reasons for your answer.
- b) Model the described problem as a (mixed) integer linear program. If the number of necessary constraints is very large, describe them compactly.
- c) What changes for the model and program if employees are only asked to leave for a trip once a week?

Self-study Exercises

Exercise S9.1. *Two Travelers*

You intend to establish a delivery service in Garching. You have two delivery vans at your disposal for this purpose. You want to deliver to your customers in such a way that you incur minimal transportation costs. Given an undirected complete graph $G = (V, E)$ with cost function $c : E \rightarrow \mathbb{R}_{\geq 0}$, where each node represents one customer and $c_{ij} := c(\{i, j\})$ models the travel cost from customer i to customer j . Your warehouse is located at node $1 \in V$. This is where the tours of the delivery trucks start and end. Of course, each customer must be on the tour of one of your delivery trucks.

- a) Give a graph with at least 3 nodes and a cost function such that it is cheaper to use only one of the delivery carts.
- b) Give a graph and a cost function such that it is optimal to visit a node (except 1) more than once. Does your solution satisfy the triangle inequality?
- c) You now additionally require that the cases in (a) and (b) must not occur, i.e., each delivery truck must make a true round trip over a subset of the nodes and each node except 1 must be visited exactly once in total. Formulate an IP to solve the problem.

Exercise S9.2. *Slitherlink*

In the logic puzzle Slitherlink, an $n \times m$ point grid is given. Two points are adjacent if they are directly next to or above/below each other in the grid. The goal is to connect adjacent points with edges, under the following conditions:

- Diagonal edges are not allowed. Only points that are directly next to or above/below each other can be connected.
- At the end, a single closed loop should be formed.
- The grid contains additional 'constraint numbers' between 0 and 3, which are located between 4 grid points. These indicate how many edges must exist between these 4 points.

Consider the following example and a corresponding solution.

- a) Model Slitherlink as an ILP, but initially leave out the subtour elimination constraints. Why can't you use the standard condition from the lecture?
- b) The following alternative for modeling subtour elimination constraints with real-valued variables $u_i \geq 0$ for the classic TSP goes back to Miller, Tucker, and Zemlin:

$$\begin{aligned}
 u_1 &= 1 \\
 u_i &\geq 2 && \text{for } i \in V \setminus \{1\} \\
 u_i &\leq n && \text{for } i \in V \\
 u_i - u_j + 1 &\leq (n-1)(1 - x_{ij}) && \text{für } i, j \in V, j \neq 1
 \end{aligned}$$

Explain how these constraints eliminate subtours. Under what conditions can you use these constraints in Slitherlink as well?

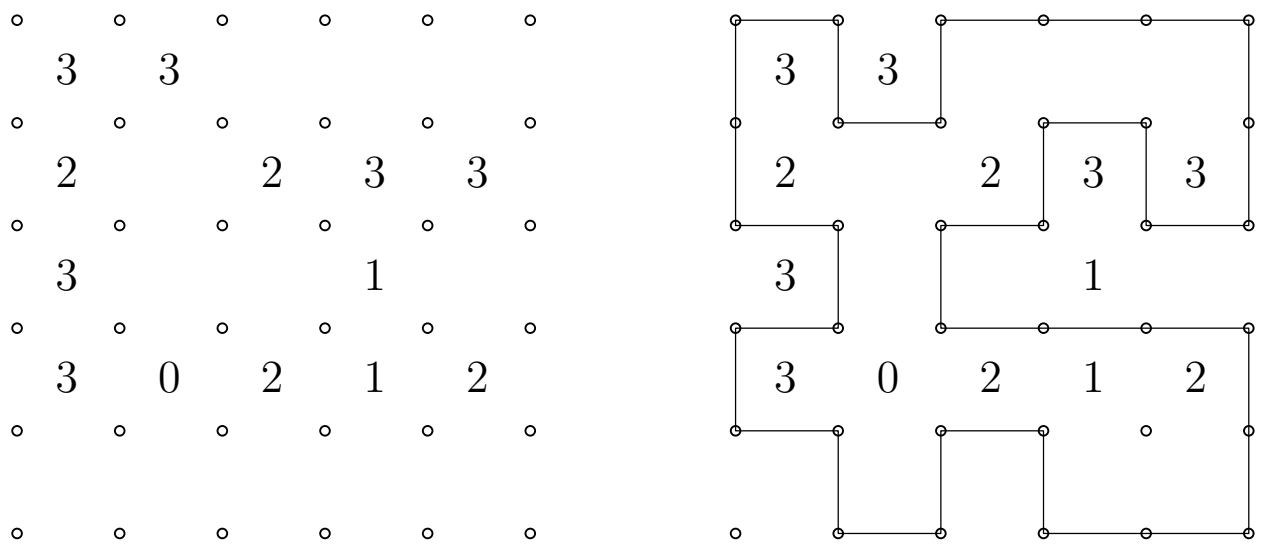


Figure 1: Slitherlink puzzle with a corresponding solution